PALANTIR C2

AI-Powered Command & Control

Automating the F2T2EA Kill Chain with Multi-Agent AI,
Drone Swarm Orchestration & Real-Time Sensor Fusion

THE PROBLEM

Modern battlespaces overwhelm human decision-makers

Information Overload

Operators face hundreds of sensor feeds, tracks, and intelligence reports simultaneously across multiple domains

Speed of Decision

Adversary kill chains compress timelines from hours to minutes — human-only loops cannot keep pace

Sensor Silos

EO/IR, SAR, SIGINT, and GMTI data remain fragmented across systems, preventing unified threat pictures

Accountability Gap

Autonomous systems lack transparency; commanders need auditable, explainable AI recommendations with HITL gates

OUR SOLUTION

End-to-end AI-assisted kill chain automation

FIND ▶ FIX ▶ TRACK ▶ TARGET ▶ ENGAGE ▶ ASSESS

- 9 AI Agents orchestrate each phase autonomously
- Multi-sensor fusion across EO/IR, SAR, SIGINT, GMTI
- Two-gate Human-in-the-Loop approval for every strike
- 10Hz real-time simulation with physics-based UAV dynamics

SYSTEM ARCHITECTURE

Four tightly integrated subsystems

FastAPI Backend

10Hz WebSocket loop
Dual-client model
3 autonomy levels
Intel feed routing

Simulation Engine

Physics-based UAV dynamics
Verification state machine
Sensor fusion pipeline
Swarm coordination

React + Cesium Frontend

3D geospatial globe
6 map modes & overlays
Multi-layout drone cameras
75 UI components

AI Agent Layer

9 LangGraph agents
F2T2EA kill chain pipeline
5 support agents
Multi-LLM fallback

AI AGENT ORCHESTRA

Kill Chain Pipeline — 4 Primary Agents

01

ISR Observer

Multi-domain sensor fusion across UAV, satellite, and SIGINT feeds. Track correlation with confidence scoring and retasking recommendations.

02

Strategy Analyst

Evaluates each detection against theater-specific Rules of Engagement. Produces priority scores and nomination recommendations.

03

Tactical Planner

Generates three ranked Courses of Action optimized for speed, probability of kill, and cost efficiency. Human selects preferred COA.

04

Effectors Agent

Executes authorized strikes with probabilistic engagement simulation. Conducts post-strike Battle Damage Assessment and reports outcomes.

FIND → FIX → TRACK → TARGET → ENGAGE → ASSESS

AI AGENT ORCHESTRA

5 Specialist Support Agents

Pattern Analyzer	Predictive intelligence — detects anomalies in convoy frequency, new facilities, and movement surges across historical data
AI Tasking Manager	Optimizes sensor retasking — evaluates asset availability and directs nearest high-fidelity assets for verification
Battlespace Manager	Geospatial backbone — threat-ring modeling, coverage gap detection, and terrain-aware mission path generation
Synthesis Query Agent	Natural language interface — converts raw tactical data into executive SITREPs for commanding officers
Performance Auditor	Process improvement — audits kill-chain outcomes, flags discrepancies, detects model drift across engagements

All agents run in heuristic mode by default (no API keys required). With LLM keys, agents upgrade via multi-provider fallback: Gemini → Anthropic → Ollama → Heuristic.

MULTI-SENSOR FUSION

Complementary fusion across all sensor modalities

$$\text{Confidence} = 1 - \text{PRODUCT}(1 - c_i)$$

EO/IR
Electro-Optical & Infrared imaging

SAR
Synthetic Aperture Radar

SIGINT
Signals Intelligence & RF spectrum

GMTI
Ground Moving Target Indicator

FMV
Full Motion Video

- Max-within-type deduplication prevents sensor redundancy
- Unscented Kalman Filter (UKF) per-target with temporal decay
- Cross-sensor disagreement detection flags conflicting intelligence
- Immutable FusionResult dataclasses ensure data integrity

SWARM COORDINATION

Intelligent multi-UAV task assignment

11 **UAV Flight Modes**

SEARCH, FOLLOW, PAINT, INTERCEPT, SUPPORT, VERIFY, OVERWATCH, BDA, IDLE, REPOSITIONING, RTB

120s **Task TTL**

Automatic task expiry with target state transition release

10Hz **Update Rate**

Physics-based simulation with smooth heading arcs

Coordination Engine

- Greedy UAV-to-target assignment with sensor-gap detection
- Priority scoring: threat weight × verification gap × coverage need
- Idle-count guard ensures reserves for new detections
- Formation types auto-adjust per target type and sensor loadout
- Zone-based imbalance tracking drives autonomous repositioning
- Automatic release on target state transitions

HUMAN-IN-THE-LOOP

Two-gate approval system for every engagement

GATE 1: NOMINATION

ISR Observer + Strategy Analyst propose targets based on sensor fusion and ROE evaluation.

Operator actions:

- APPROVE — target advances to COA generation
- REJECT — target removed from strike pipeline
- RETASK — request additional sensor verification

GATE 2: COA AUTHORIZATION

Tactical Planner generates 3 ranked Courses of Action, each optimized for a different priority.

Ranked by:

- Speed — fastest time-to-effect
- P(Kill) — highest probability of success
- Cost — most resource-efficient option

Three Autonomy Levels

MANUAL

All decisions require operator input

SUPERVISED

AI proposes, operator approves/rejects

AUTONOMOUS

AI executes with operator monitoring

OPERATIONAL INTERFACE

Professional 3D geospatial command dashboard

Cesium 3D Globe

Real-time entity rendering with 19 dedicated hooks for drones, targets, zones, and overlays

6 Map Modes

OPERATIONAL, COVERAGE, THREAT, FUSION, SWARM, and TERRAIN views with keyboard shortcuts

Multi-Layout Cameras

SINGLE, PIP, SPLIT, and QUAD drone camera layouts with 4 sensor modes (EO/IR, SAR, SIGINT, FUSION)

4 Sidebar Tabs

MISSION (theater, intel, strikes), ASSETS (drone control), ENEMIES (targets, verification), ASSESS (threats)

SIGINT Display

RF spectrum waterfall visualization with real-time signal classification and emitter geolocation

5 Layer Overlays

Coverage, Threat, Fusion, Swarm, and Terrain overlays with per-layer visibility toggles

THEATER CONFIGURATIONS

Pre-configured for diverse operational scenarios

Romania Eastern Flank

NATO Defensive

BLUE FORCE

20 MQ-9 UAVs (24h endurance)

RED FORCE

SAM, TEL, TRUCK, CP, MANPADS, RADAR, C2, LOGISTICS

TERRAIN

Grassland / Clear weather

South China Sea

Maritime Patrol

BLUE FORCE

15 P-8 Poseidon (10h endurance)

RED FORCE

DESTROYER, FRIGATE, MISSILE BOAT, SUBMARINE, SAM, RADAR

TERRAIN

Maritime / Partly cloudy

Baltic States Defense

NATO Forward Defense

BLUE FORCE

12 MQ-9 UAVs (20h endurance)

RED FORCE

SAM, TEL, MANPADS, RADAR, APC, IFV, MBT, C2, LOGISTICS

TERRAIN

Urban-forest / Overcast

Theaters are YAML-configurable — add custom scenarios in minutes with custom Blue/Red ORBAT, weather, terrain, and ROE.

TARGET VERIFICATION

Rigorous state machine prevents false engagements

DETECTED ▶ CLASSIFIED ▶ VERIFIED ▶ NOMINATED

Threat-Adaptive Thresholds

High-threat targets (SAM, TEL, MANPADS) advance through verification faster due to urgency

Low-priority targets (TRUCK, LOGISTICS) require higher confidence thresholds before nomination

Per-target-type configurable thresholds ensure proportional response

DEMO_FAST preset halves all verification times for live demonstrations

Safety Mechanisms

Regression timeouts prevent stale verifications from lingering in the pipeline

ROE identity classification: HOSTILE, SUSPECT, NEUTRAL, FRIENDLY

Theater-specific Rules of Engagement loaded from YAML configs

Pure-function state machine ensures deterministic, testable behavior

BATTLESPACE ASSESSMENT

Real-time tactical intelligence at a glance

Threat Clustering

Automatic grouping of hostile assets into SAM_BATTERY, CONVOY, CP_COMPLEX, AD_NETWORK, and MIXED clusters for prioritized engagement

Coverage Gap Detection

Identifies under-surveilled zones and triggers autonomous UAV repositioning to eliminate blind spots

Zone Threat Scoring

Per-zone threat level computation based on target types, count, and proximity — drives resource allocation

Movement Corridors

Predictive analysis detects likely movement routes for shoot-and-scoot TELs and convoy assets

BY THE NUMBERS

Production-grade engineering metrics

475
Pytest Tests
Across 23 test files

138
Python Modules
32,000+ lines of code

75
React Components
TypeScript + Blueprint

9
AI Agents
LangGraph orchestrated

30
WebSocket Actions
Real-time bidirectional

10
Target Types
With unique behaviors

USE CASES

Deployable across defense and security domains

Air Defense Suppression (SEAD/DEAD)

Automated detection, tracking, and engagement of SAM sites and mobile TEL launchers across contested airspace

Maritime Domain Awareness

Multi-sensor ISR across South China Sea with P-8 Poseidon fleet tracking destroyers, submarines, and fishing vessels

Forward Area Air Defense

NATO Eastern Flank defense with MQ-9 swarms coordinating against armored columns, IFVs, and MANPADS threats

Counter-UAS Operations

Detection and intercept of enemy RECON, ATTACK, and JAMMING UAVs with dedicated intercept modes and swarm response

Intelligence Preparation

Pattern analysis across historical data to predict adversary movements, identify new facilities, and detect operational surges

Training & Simulation

Full-fidelity tactical simulator with physics-based dynamics for operator training without live assets or real-world risk

TECHNOLOGY STACK

Built on battle-tested open-source foundations

BACKEND

- Python 3.11+
- FastAPI + WebSocket
- LangGraph / LangChain
- Pydantic Models
- Async JSONL Logging
- python-dotenv

FRONTEND

- React + TypeScript
- Vite Build System
- CesiumJS 3D Globe
- Blueprint UI (Dark)
- Zustand State Mgmt
- Canvas Rendering

AI / ML

- Multi-LLM Fallback
- Gemini / Anthropic / Ollama
- Heuristic Default
- Unscented Kalman Filter
- Bayesian Fusion
- Pattern Recognition

DEMO MODE

Full kill chain in auto-pilot — zero human input

```
$ ./palantir.sh --demo
```

0s	System launches with Romania theater, 20 UAVs begin SEARCH patrols
~30s	Sensors detect and classify SAM sites, TELs, and mobile targets
~60s	Verification engine promotes high-confidence targets to NOMINATED
+5s	Auto-pilot APPROVES nominations (Gate 1) after configurable delay
+3s	Tactical Planner generates 3 COAs, auto-pilot AUTHORIZES highest-scored
+10s	Effectors Agent executes strike — probabilistic hit/kill simulation
+5s	BDA assessment begins — UAV repositions for damage observation

WHY PALANTIR C2

Competitive advantages that set us apart

⭐ End-to-End Automation

Only system that covers the complete F2T2EA kill chain from sensor-to-shooter with AI orchestration at every stage

⭐ Auditable AI Decisions

Two-gate HITL, ROE evaluation, and async JSONL logging create a full audit trail for every recommendation and engagement

⭐ Sensor-Agnostic Fusion

Complementary fusion formula works across any sensor type — not locked to specific hardware or data formats

⭐ Zero-Config Operation

Runs in heuristic mode without any API keys. LLM integration is optional — upgrades intelligence when available

⭐ Theater Portability

YAML-configurable theaters with custom ORBAT, ROE, weather, and terrain. New scenarios deployable in minutes

⭐ Production-Grade Quality

475 tests, typed data contracts (Pydantic + TypeScript), structured logging, and comprehensive error handling

PALANTIR C2

Accelerate decisions. Dominate the battlespace.

REQUEST A LIVE DEMO